

Towards Conducting Effective Locomotion Through Hardware Transformation in Head-Mounted-Device - A Review Study

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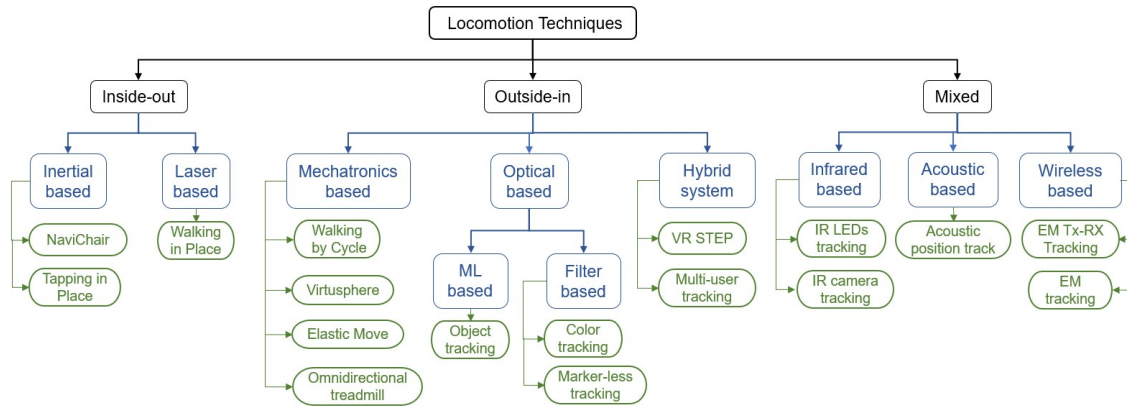


Figure 1: Taxonomy of the locomotion techniques based on motion tracking hardware

ABSTRACT

Locomotion in Virtual Reality (VR) acts as a motion tracking unit for simulating user movements based on the Degree-of-Freedom (DOF) of the application. For effective locomotion, VR practitioners may have to transform their hardware from 3-DOF to 6-DOF. In this context, we conducted a literature review on different motion tracking methods employed in the Head-Mounted-Devices (HMD) to understand such hardware transformation to conduct locomotion in VR. Our observations led us to formulate a taxonomy of the tracking methods for locomotion in VR based on system design. Our study also captures different metrics that VR practitioners use to evaluate the hardware based on the context, performance, and significance for conducting locomotion.

Index Terms: Hardware—Communication hardware, interfaces and storage—Sensor devices and platforms—

1 MOTIVATION

VR products are on the rise. With FacebookTM announcing its vision for metaverse¹, there may be going to be a storm of VR applications into the market like never before. For every VR product, locomotion plays a crucial role in engaging the participant with the VR content. Effective locomotion is achieved if suitable motion tracking methods are best utilized. Unlike external haptic devices, HMDs also offer

various avenues to conduct motion tracking to achieve various types of locomotion. However, locomotion capabilities vary between HMDs with a 3-DOF and a 6-DOF. Usually, VR applications that require 3-DOF support can be executed on a 6-DOF supported HMD. However, a VR application that requires 6-DOF support may not be executed on a 3-DOF supported HMD. In contrast, by facilitating some additional hardware on a 3-DOF supported HMD, a closer 6-DOF application can be executed. Such practice is prevalent in the VR practitioner community as most of them upscale or downscale the HMD capacity on supporting motion tracking to manage effective locomotion. Considering these facts, we conducted a literature review on available literature to examine the practices adopted by VR practitioners on how the HMD hardware was transformed from 3-DOF to 6-DOF motion tracking for effective locomotion.

2 REVIEW STUDY STEP

This study helps us understand the hardware transformation of HMDs from 3-DOF to 6-DOF without additional external haptic support for conducting locomotion. To achieve this, we conducted a systematic literature study by considering the guidelines proposed by Kitchenham et al. [1]. We utilized the PICOC (Population, Intervention, Comparison, Outcome, and Context) method [2] to establish our study's context and relevance [3]. We aim to summarize the motion tracking methods served by HMDs for conducting effective locomotion in VR applications. Below research questions are expressed to capture the insights of our objective.

- **RQ1**-What types of motion tracking methods are operated for head tracking by an HMD for VR applications?
- **RQ2**: What are the hardware components required for transforming a VR HMD from 3-DOF to 6-DOF motion tracking?
- **RQ3**: What are the metrics practiced to evaluate the effectiveness of motion tracking methods after transforming the VR HMD from 3-DOF to 6-DOF?

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¹<https://tech.fb.com/connect-2021-our-vision-for-the-metaverse/>

After a thorough search quality assessment as illustrated in our resources [3], we deduce our search string as follows. Overall our search string is defined S_1 AND S_2 AND S_3 .

S_1 : "Virtual Reality" OR "VR"

S_2 : "Head mounted device" OR "Head mounted display" OR "HMD" OR "Head-mounted display" OR "Head-mounted device" OR "headset" OR "display" OR "projection"

S_3 : "Degree of Freedom" OR "Degrees of Freedom" OR "DOF" OR "3 DOF" OR "3-DOF" OR "3DOF" OR "6 DOF" OR "6-DOF" OR "6DOF" OR "motion tracking" OR "motion-tracking" OR "head tracking" OR "head-tracking"

We conducted multiple iterations to arrive at a finalized search string. Our search interval is between 2011 - 2021 as the publications before 2010 did not provide any significant research contribution. Our review study includes papers published until August 2021. Our search results returned 3009 papers (ACM:998, IEEE: 726, Springer: 1285). Using our inclusion and exclusion criteria, the first iteration screening returned 767 papers. Further snowballing [4] on content check resulted in 17 papers. The output seem to be too low as there is very little literature on HMD hardware transformation from 3-DOF to 6-DOF for conducting locomotion in VR. All the authors have arrived at a similar conclusion towards the search results. All the supplementary material [3] of our search iterations and review are made available for replicating our study.

3 OBSERVATIONS

In this section, we discuss our insights in regards to respective research questions as follows:

RQ1-What types of motion tracking methods are operated for head tracking by an HMD for VR applications? - Primary Tracking methods practiced as color tracking, VR-STEP, Walking by Cycle, Omni-irectional treadmill, object tracking, Suspended Walking, Walking In-place, Infrared Tracking method, Multi-user tracking, Nav-Chair, Virtusphere, Tapping In-place, Electromagnetic Tracking System, Elastic-Move and Acoustic Position Tracking method. Rotational, Axial and Translation based locomotion methods are employed using the observed tracking methods. They are used as VR based applications like Sports, Cognitive Exploratory studies, Redirectional Motion, Multi-User based training and simulation studies. The working principle, hardware setup of tracking method and its related locomotion are illustrated as part of our resources [3].

RQ2: What are the hardware components required for transforming a VR HMD from 3-DOF to 6-DOF motion tracking? - we address this research question by proposing a taxonomy of locomotion techniques based on hardware support as shown in Figure 1. The underlying hardware is used to conduct locomotion through motion tracking methods in a given HMD. The motive behind illustrating the taxonomy is to help VR practitioners establish a relationship between the hardware needs of an HMD and choose a suitable locomotion method. As shown in Fig 1, the boxes in blue are types of respective locomotion techniques categorized as Inside-out, Outside-In, and Mixed. The boxes in green are the instances or examples of these respective types listed in blue boxes. **Inside-Out tracking** - The sensors are located on the hardware (HMD) or the user peripherals as part of the Inside-Out tracking. **Outside-In tracking** - The sensors are placed externally, preferably in a stationary position, and are not administered directly on the user's device as part of Outside-In tracking. **Mixed tracking** - Further studies have found to be following mixed-methods i.e., they employ both Inside-Out and Outside-In tracking. We categorized them as Mixed Tracking methods.

RQ3: What are the metrics practiced to evaluate the effectiveness of motion tracking methods after transforming the VR HMD from 3-DOF to 6-DOF? - The majority of the metrics are used for studies involving focused groups, Simulation, Comparative, User-Centered, and Cognitive experiments. These metrics are Position and Rotation Accuracy, Drift Measure, Track resolution, Switching rate, Portability and Precision, Noise Immunity, Power Analysis, Force and Response time, System and Statistical analysis, Motion-to-Photon latency. Detailed mapping of relevant metric with the study type are illustrated as part of our supplement material [3]. Details on tracking method with respective working principle employed using a specific hardware setup is illustrated in Table as part of resources [3].

4 CONCLUSION AND FUTURE WORK

Many motion tracking techniques were implemented based on actions like walking, steering, selection and manipulation [5] [6] [7]. However, these studies on VR locomotion do not discuss the device parameters, metrics for evaluation and the hardware requirements for designing a particular motion tracking system. Device's operating range, sensor ability, and its relation with target application are essential viewpoints for potential research. We tried to address such need as part of our review by proposing a novel taxonomy for locomotion. Overall, we expect that our review study will help HMD developers to consider all available observations to build novel HMDs using distinct motion tracking methods to conduct locomotion for focused VR applications. This review helped us channelize the strengths of prevailing hardware transformation setups, evaluation practices, and potential metrics practiced by VR practitioners. We plan to use this review study to develop a customized motion tracking method in a novel 3-DOF HMD for a focused healthcare VR applications. These healthcare VR applications are planned to be low-cost and have better ease of use.

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